

A KoBERT–SVD Hybrid Model for Personalized Cosmetic Recommendations Based on User Sentiment and Skin Attributes

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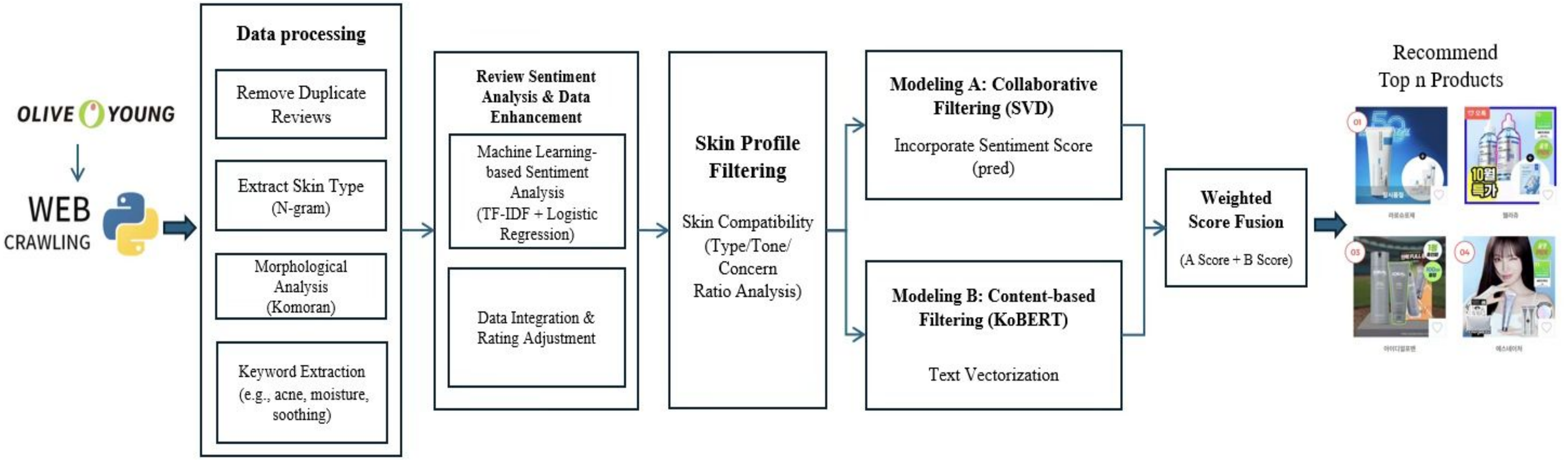
Outline

1. Research Framework: SIX STEPS
2. Final Recommendation Results
3. Limitations and Future Research Directions

Background

AI Growth	Rapid growth of AI → Data-driven personalization
Skin Diversity	High diversity in skin types and tones
Current Limits	Current systems = limited (ratings or history only)
Missing Elements	Lack of emotional & skin-based understanding
Research Goal	Build hybrid model reflecting emotion and skin profile

Research Overview: Six Stages



Data & Preprocessing

12,347
product reviews collected

—

522
duplicates removed

=

11,795
valid dataset extracted

Data Collection

Web Crawling of Cosmetic Review Data from the Olive Young Website

Duplicate Removal

Removal of Identical Reviews and Duplicate User Entries

Skin Type Extraction

Extraction of Skin Type, Tone, and Concerns Using the N-gram Technique

Morphological Analysis

Keyword Extraction Using the Komoran Morphological Analyzer

Table 1. Data columns and descriptions

Name	*Detail*
item_id	Product data
user_id	User ID
skin_type	Skin type (dry, oily, etc.)
skin_tone	Brightness, darkness, etc.
skin_concerns	Sensitivity, dehydration, acne, etc.
review	User comments
rating	Score(1~5)

Sentiment Analysis

Sentiment Analysis Using TF-IDF and Logistic Regression



Review Text

User Comments and Original Ratings





Sentiment Analysis

Rule-Based Approach Using Sentiment Lexicon TF-IDF Vectorization and Logistic Regression Model Application

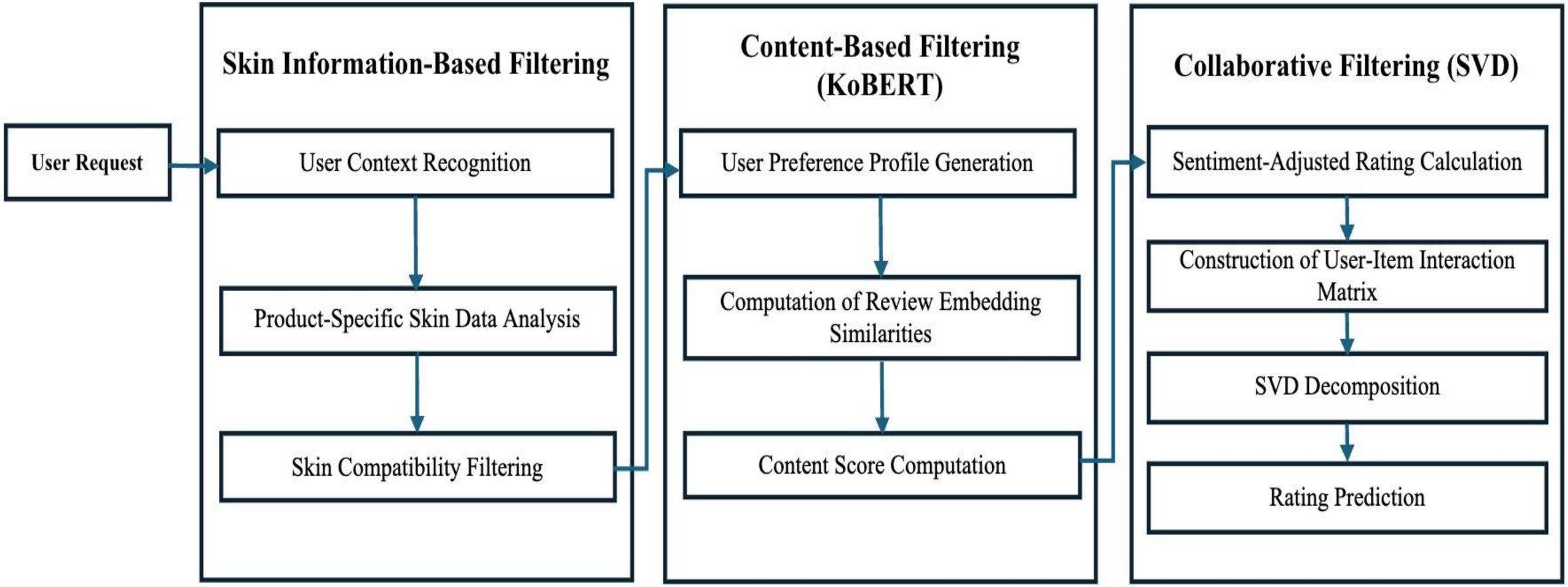




Rating Augmentation

Integrating Sentiment Score (Positive/Negative) into Original Ratings Created “rating_aug” to Reflect Users’ Emotional Satisfaction

Skin Profile Filtering

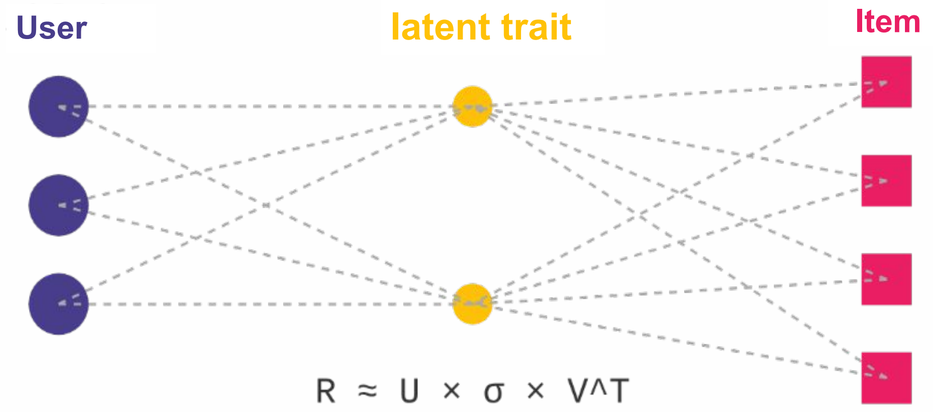


Model A: SVD Collaborative Filtering

SVD Model Implementation

- Utilizing Sentiment-Adjusted Ratings
- User–Item Interaction Matrix
- Matrix Factorization

Visualization of the SVD Model Concept



$$A=U\Sigma V^T$$

	SVD	*KNN*	*NMF*
RMSE	0.5481	0.5717	0.6594
MAE	0.3095	0.3217	0.4418

Model B: KoBERT Content Filtering

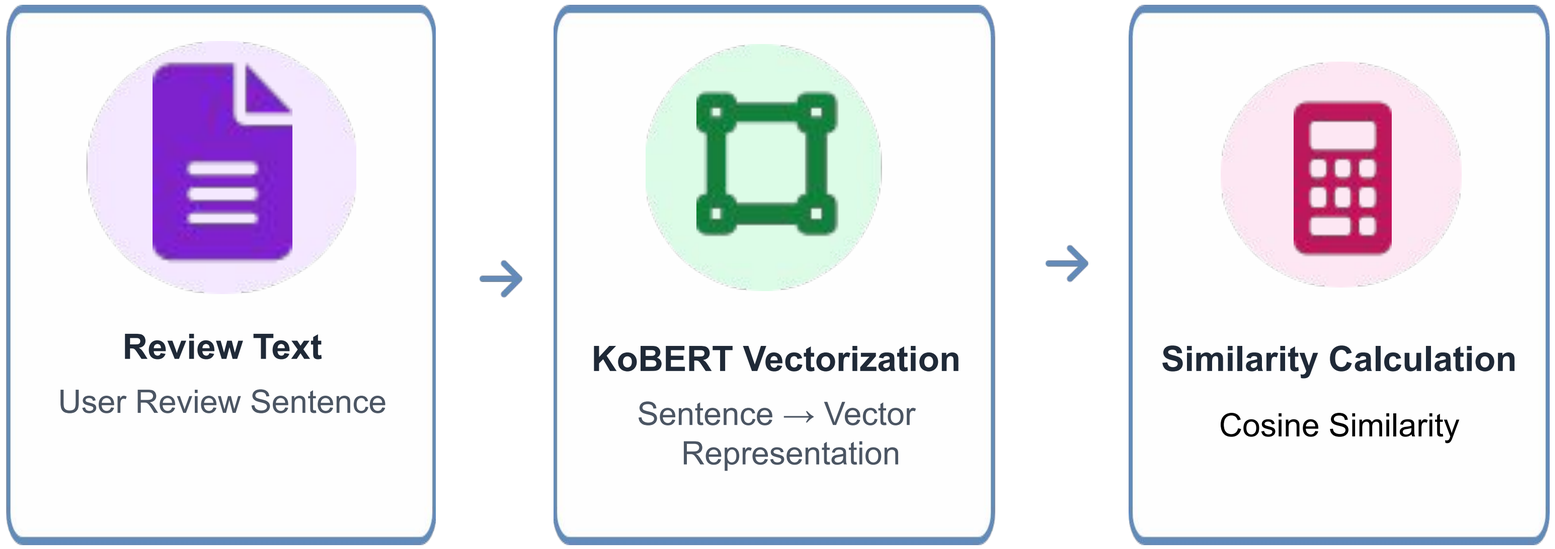
What is KoBERT?

- A Korean Language–Specialized Model (Korean BERT)
- Captures the Contextual Meaning of Review Texts
- Used for Generating User Preference Profiles

Model Objective

- Provides Meaning-Based Recommendations Beyond Simple Keyword Matching
- Calculates Contextual Similarity of Review Texts

KoBERT Content Filtering Process



Hybrid Integration

Model A (Collaborative Filtering)

SVD-Based Recommendation Score

Model B (Content-Based Filtering)

KoBERT-Based Recommendation Score

Weighted Score Fusion

- SVD score : w_1, KoBERT score : w_2
- Final Score = $w_1 \times SVD_Score + w_2 \times KoBERT_Score$
- Combining the SVD Model and the KoBERT Model

Final Recommendation

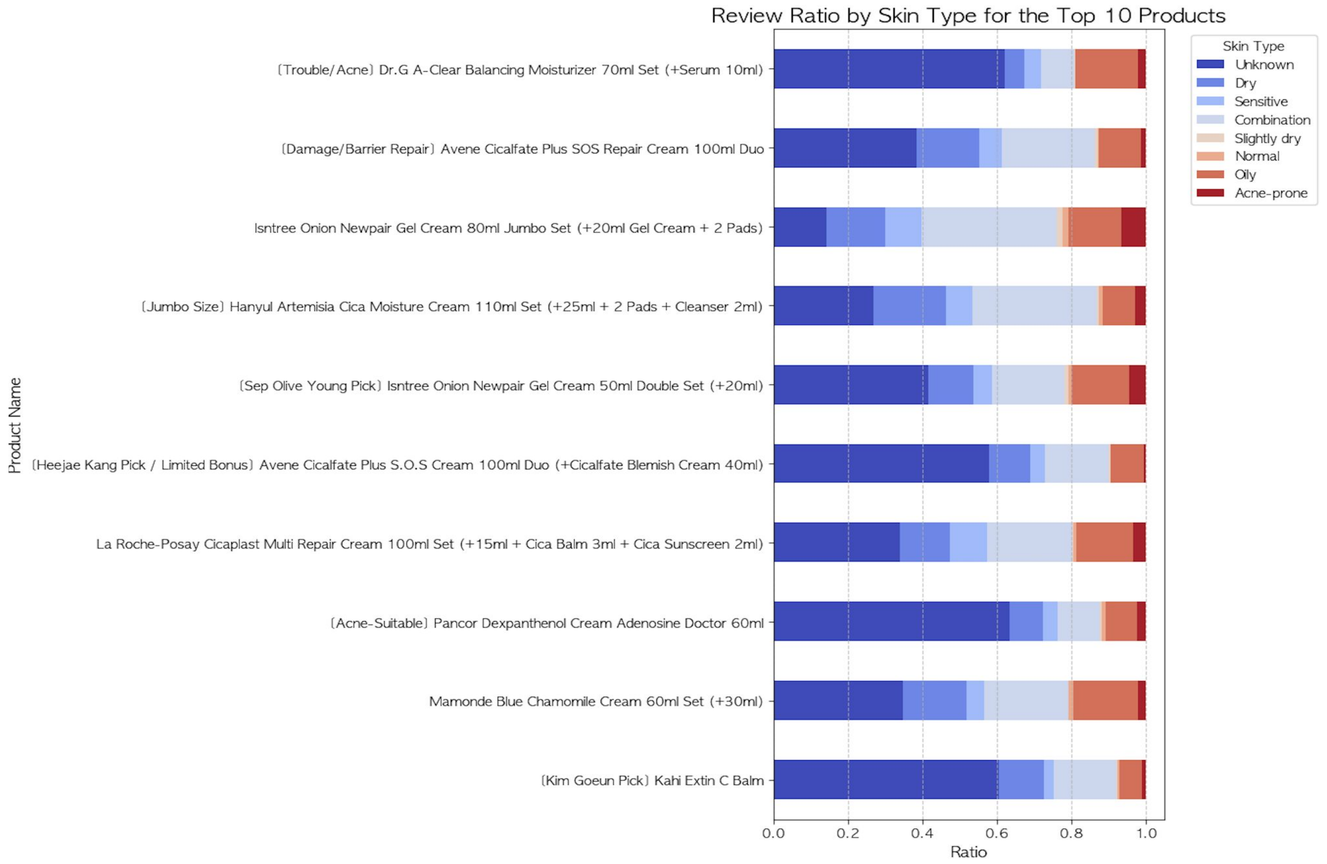
Top N Products Recommended

Results & Evaluation

Top-n items

	item_id	final_score
0	마몽드 블루 캐모마일 크림 60ml 기획 (+30ml)	0.879244
1	[아랑 PICK] 낫츠 센텔라스카 연고 일랑일랑 15g	0.832843
2	[손상/장벽 리페어] 아벤느 시칼파트 플러스 SOS 리페어 크림 100ml 2입 기획	0.752454
3	이즈앤트리 어니언 뉴페어 겔크림 80ml 대용량 기획 (+겔크림20ml+패드2매)	0.726204
4	[응급진정/38% 여드름 개선] 로벡틴 시카케어 블레미쉬 클리어링 크림 50ml 리...	0.654759
5	[강희재픽/추가증정한정기획] 아벤느 시칼파트 플러스 S.O.S 크림 100ml 2입...	0.621607
6	키엘 울트라 쉐이셜 크림 28ml	0.599900
7	[대용량기획] 한율 쑥시카 수분크림 110ml (+25ml+ 패드2매+흡착팩폼2ml)	0.578260
8	키엘 울트라 쉐이셜 오일-프리 젤 크림 50ml	0.400000
9	[김고은 Pick] 가히 엑스틴C밤	0.362142

Proportion of reviews for each product



Conclusion

Expected Outcomes

1. Personalization Enhancement
→ Improves recommendation accuracy by reflecting both skin characteristics and user emotions.

2. Emotion-Based Recommendation
→ Provides customized results by analyzing positive and negative sentiments from user reviews.

3. Efficient Computation
→ Reduces computational load and increases processing speed through skin profile filtering.

Future Research Directions

1. Integration of Multimodal Data Sources
→ Incorporate ingredient composition, dermatological test results, and real-time user feedback.

2. User Study Validation
→ Conduct A/B testing and real-user evaluations to verify performance and satisfaction.

3. Model Generalization Enhancement
→ Expand model applicability across various skin types, age groups, and cultural backgrounds.



Conclusion & Next Steps

This study proposes a novel SVD-KoBERT hybrid model
for personalized cosmetic recommendations.

It effectively integrates collaborative and content-based filtering
by analyzing user sentiment and skin attributes to deliver more accurate results.

Thank You

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